

Logistics App Database Analysis

Entities:

- User
- Role
- Permission
- Role_Permission
- Client
- Branch
- Driver
- Car
- Order
- Order_Status_History
- Order_Images
- Car_Maintenance_History
- Car_Status_History
- Trip (Daily Report / Shift)

Entities Attributes :

User:

- id
- name
- email
- password
- role_id
- branch_id
- created_at
- updated_at

Role:

- id
- name
- created_at
- updated_at

Permission

- id
- name
- slug
- module
- created_at
- updated_at

Role_Permission:

- id
- role_id
- permission_id

Client:

- id
- name
- phone
- email
- address
- client_type
- created_at
- updated_at

Branch:

- id
- name
- created_at
- updated_at

Driver:

- id
- name
- address
- nationality
- national_id
- national_id_type
- gosi_number
- phone
- license_number
- license_type
- license_expiry
- photo
- driver_card_number
- driver_card_type
- driver_card_expiry
- driver_type
- status` (Enum: Active, Inactive, OnTrip)
- created_at
- branch_id
- national_photo
- driver_card_photo
- updated_at

Car:

- id
- manufacturer
- model
- year
- color
- plate_number
- plate_letters
- plate_type
- registration_number

- vin_number
- owner_national_id
- actual_user_id
- actual_user_start_date
- branch_id
- ownership_start_date
- registration_issue_date
- registration_expiry_date
- insurance_status
- insurance_expiry_date
- inspection_status
- inspection_expiry_date
- capacity
- weight
- registration_photo
- insurance_photo
- operation_card_number
- operation_card_expiry
- gps_device_id
- current_status
- created_at
- updated_at

Car_Maintenance_History:

- id
- car_id
- date
- reason
- cost
- start_at
- end_at
- created_at

Car_Status_History:

- id
- car_status
- car_id
- impound_status
- created_at

Order:

- id
- shipment_number
- sender_name
- recipient_name
- recipient_phone
- pickup_building_no
- pickup_street
- pickup_district
- pickup_city
- pickup_zip
- pickup_additional_no
- pickup_unit_no
- delivery_building_no
- delivery_street
- delivery_district
- delivery_city
- delivery_zip
- delivery_additional_no
- delivery_unit_no
- is_national_address_validated
- sub_total
- vat_rate`
- vat_amount
- total_price
- payment_method
- payment_status
- tax_invoice_number
- current_status
- type
- quantity
- weight
- trip_id

- client_id
- created_at
- updated_at

Order_Status_History:

- id
- order_id
- status
- reason
- created_at

Order_Image:

- id
- order_id
- image_url
- public_id
- created_at

Trip:

- id
- driver_id (FK)
- car_id (FK)
- start_time
- end_time
- collected_count
- delivered_count
- returned_count
- total_cash_collected
- end_reason
- notes
- status
- created_at
- updated_at

Relationships:

User - Branch:

- Each user belongs to one branch (N:1 relationship).
- Foreign Key: User.branch_id references Branch.id.

User - Role:

- Each user is assigned to one role (N:1 relationship).
- Foreign Key: User.role_id references Role.id.

Role - Permission (M:N via Role_Permission):

- Roles have many permissions, and permissions belong to many roles.
- Foreign Keys: Role_Permission.role_id, Role_Permission.permission_id

Driver - Branch:

- Each driver belongs to one branch (N:1 relationship).
- Foreign Key: Driver.branch_id references Branch.id.

Car - Branch:

- Each car is assigned to one branch (N:1 relationship).
- Foreign Key: Car.branch_id references Branch.id.

Car - Car_Maintenance:

- Each car can have multiple maintenance records (1:N relationship).
- Foreign Key: CarMaintenance.car_id references Car.id.

Car - Car_Status_History:

- Each car can have multiple status change events (1:N relationship).
- Foreign Key: Car_Status_History.car_id references Car.id.

Order - Order_Status_History:

- Each order can have multiple status updates (1:N relationship).
- Foreign Key: OrderStatus.order_id references Order.id.

Order - Order_Image:

- Each order can have multiple images (1:N relationship).
- Foreign Key: OrderImage.order_id references Order.id.

Trip - Order

- One Trip has many Orders (1:N)
- Foreign Key: Order.trip_id references Trip.id

Trip - Driver:

- One driver can have many trips/shifts over time (1:N relationship).
- Foreign Key: Trip.driver_id references [Driver.id](#).

Trip - Car:

- One car can be used in many trips/shifts over time (1:N relationship).
- Foreign Key: Trip.car_id references Car.id.

Client - Order:

- One Client can have many Orders (1:N relationship).
- Foreign Key: Order.client_id references Client.id.